

Where chip-supply rent migrates next: eight inelastic nodes, six that survive

The binding constraint is rarely the named monopoly. It is the artisanal sub-step, the byproduct metal, the second-decade engineer, and the isotope cascade that money cannot ramp on a fab timeline.

Thesis spine: Frontier → Capability → Dependency graph → Supply elasticity → Demand → Capital → Pricing → Policy → Outcomes. Rent accrues to the inelastic input. The edge is naming where the constraint moves before pricing catches up.

Area: semiconductors and the chip supply chain (logic foundry, memory and HBM, advanced packaging and substrates, lithography and fab equipment and specialty gases/materials, EDA and IP, test, and the compute hardware stack) • Horizon: 2030 to 2040

Method: generate wide and disruptive, then gate strict. Each call names the needle, not the theme.

Two probabilities per call: directional vision, and the strict dated clause scored at resolution.

Status: hardened candidates (survived the adversarial refute pass). Drafted 2026-06-14.

The board: 6 hardened calls

The cross-cutting read

These six calls share one structural shape: capital, tools, and wafer capacity are all financializable on a capex clock, but the actual ceiling sits one or two layers below the priced narrative at an input whose supply grows on a physical or biological clock, not a financial one. Three of the six are tacit-knowledge or labor constraints (Zeiss Oberkochen optics figuring, senior yield/integration engineers, certified UHP hookup welders) where the supply curve is set by how fast a human accumulates a decade of on-line experience or a trade certifies a fitter, neither of which parallelizes with money. Two are single-vendor or single-site equipment chokepoints (Zeiss again, IMS multibeam mask writers) where lead times of 18 to 36 months and a nonexistent second qualified source cap how fast output can rise. Two are geologically or isotopically inelastic materials (ruthenium as a 30 t/yr platinum byproduct with no primary mine, enriched Si-28 from export-controlled separation cascades). The common edge is not that these inputs are unknown, but that the equity and policy channels price the wrong unit: they price ASML the company, not Zeiss the polishing bench; the fab as capex, not the yield-curve carrier; the metal price, not the chip-ramp gating risk. The strongest calls cleanly separate vision_p (is the physical mechanism real and inelastic) from clause_p (does the specific dated, disclosure-dependent clause fire), and the lower clause scores honestly reflect substitution routes (molybdenum for ruthenium, Cu-Cu hybrid bonding for solder, superconducting qubits for silicon spin) and the fact that several resolution triggers depend on a public disclosure that supply chains may manage quietly.

At a glance

#	The boom	Binding constraint (the needle)	Vision P	Clause P	Resolves
P1	The loud narrative: ASML is the monopoly and EUV source power is the throughput knob. Capital is pouring into fab shells (TSMC AZ, Intel, Samsung, Rapidus) assuming scanners will follow. Sell-side equity models and export-control frameworks both treat the ASML scanner as the atomic unit of scarcity, without disaggregating the projection optics module as a distinct, slower-moving constraint.	Zeiss SMT deterministic ion-beam-figuring and Mo/Si multilayer-coating throughput for EUV/High-NA projection mirror sets	82%	52%	2032-12-31
P2	Capital, EUV tools, and wafer capacity are all fundable on a capex timeline; the one input	Senior (second-decade, 10-plus years) yield-enhancement and process-	82%	58%	2033-12-31

#	The boom	Binding constraint (the needle)	Vision P	Clause P	Resolves
	that is not is a human who has personally climbed the yield curve across several node generations at sub-5nm class. Process integration and yield-enhancement engineering is overwhelmingly tacit: contamination fingerprinting, defectivity root-cause, and parametric drift diagnosis are learned over 10-plus years on a running line and do not transfer from a textbook or a two-year program. The market prices a fab as steel, tools, and a node label, then assumes yield ramp follows on schedule. It does not. TSMC explicitly attributed Arizona production delays to skilled worker shortages and responded by flying in over 1,000 Taiwanese engineers on three-year assignments -- then the second Arizona plant slipped to 2027-2028. That is not a capex problem; it is a headcount-of-experienced-engineers problem. One-third of the US senior fab-engineer cohort is at or near retirement age (a fixed birth-cohort fact), degree enrollment in feeder programs lagged for years, and announced ex-Taiwan fab count is contractually committed. The mismatch between capex-speed fab buildout and biology-speed expertise accumulation is mechanical and already locked in by decisions already made.	integration engineers with multi-node hands-on ramp			
P3	Copper interconnects fail by physics at wire dimensions below roughly 12-16 nm pitch: electron surface scattering and	Ruthenium metal supply and high-purity Ru sputtering-target and ALD-precursor conversion capacity for barrierless interc	72%	40%	2035-12-31

#	The boom	Binding constraint (the needle)	Vision P	Clause P	Resolves
	grain-boundary scattering blow up resistivity, and the barrier/liner layer that copper requires eats an ever-larger fraction of the shrinking cross-section. Ruthenium is the industry-chosen escape because it can be deposited barrierless, its resistivity degrades far less at nanoscale, and it is being designed into the tightest metal levels and backside-power-delivery vias at the A14 node and below, with ramp roughly 2027-2030. The chokepoint is that ruthenium is not mined directly. It is a byproduct of platinum-group-metal extraction, roughly 30 tonnes per year globally, geologically concentrated in South Africa's Bushveld and Russia's Norilsk, with output governed by platinum and palladium demand from autocatalysts and jewelry, not chip fabs. When Ru moves from trace use in hard disks and catalysts to coating the bottom interconnect levels and backside vias of every advanced GPU, CPU and HBM die, even a few extra grams per wafer times tens of millions of leading-edge wafers collides with that 30-tonne ceiling. Rent migrates from the foundry to whoever controls Ru refining and sputtering-target conversion.				
P4	A fab is two things: a building shell and an extraordinarily complex web of ultra-high-purity gas, chemical, slurry, and process-water distribution	Certified UHP orbital welders and UHP/PVDF gas-and-chemical-distribution fitters qualified to semiconductor contaminatio	72%	52%	2031-12-31

#	The boom	Binding constraint (the needle)	Vision P	Clause P	Resolves
	that must be orbital-welded, hooked up, and certified leak-tight to SEMI particle standards before a single tool runs. That hookup work is a distinct specialty trade, certifiable only after years of pipefitting experience plus fab-specific qualification to smooth-bore, crevice-free, helium-leak-tested standards. TSMC Arizona already slipped explicitly citing this: the company had to airlift roughly 500 Taiwanese hookup specialists because the regional certified pool was too thin for even one major fab. SEMI's own data show roughly 97 high-volume fabs launching globally in 2023-2025, with the largest US/EU share in history and all on overlapping schedules. The concurrent demand is a step-function; the certified-trade pool grows only at the rate the trades can train and qualify people, which is years. Rent migrates to the specialty UHP mechanical subcontractors and to the certified orbital-welder pool; the visible symptom is schedule slip clustered on the install/hookup phase, not on construction or tool delivery.				
P5	Two independent demand curves land on one isotope-separation step. Silicon-spin qubits from Intel, Diraq, Quantum Motion and peers require 99.99%+ Si-28 to suppress Si-29 nuclear-spin decoherence -- this is non-negotiable physics, not a	Enriched Si-28 isotope-separation capacity: specifically the aerodynamic-separation (silane) or gas-centrifuge (SiF4) ca	72%	34%	2034-12-31

#	The boom	Binding constraint (the needle)	Vision P	Clause P	Resolves
	preference. Separately, removing Si-29 and Si-30 from the silicon lattice boosts thermal conductivity 60-600% over natural silicon, which matters specifically for buried-device heat extraction in CFET and backside-power-delivery 3D stacks where junction temperatures are the binding physical limit. The enrichment process -- aerodynamic separation of silane or gas-centrifuge on SiF4 -- is the same physics as uranium enrichment, so it carries export controls, long permitting timelines, and high capital intensity. ASP Isotopes (ASPI) is the only disclosed Western commercial enriched-Si-28 supplier, with a Pretoria facility restarted in May 2026 after engineering fixes, first commercial shipments targeted Q3 2026, and capacity documented at greater than 80 kg/yr from one facility. Three named purchase agreements already exist on that output (a major U.S. semiconductor company, a large industrial gases company, a large U.S. buyer). Total world capacity including any Rosatom legacy output is on the order of low hundreds of kg/yr. Any scenario where silicon-spin qubits reach even modest commercial volume, let alone where Si-28 epi gets adopted for 3D-logic thermal relief, requires 10 to 100 times that output. Aerodynamic and centrifuge cascades are among the slowest industrial plants to permit and commission;				

#	The boom	Binding constraint (the needle)	Vision P	Clause P	Resolves
	\$333M in ASPI cash begins to relax supply on a multi-year basis but does not dissolve the constraint inside a decade.				
P6	The loud narrative is the GPU and ASIC design explosion -- design talent, HBM, CoWoS packaging. Masks are treated as a solved commodity. The photomask conversation, where it exists, is about blank substrates (Hoya, AGC), not about who writes the pattern. The specialist trade press (SemiAnalysis, August 2022) has named the IMS monopoly, but the second-order mechanism -- that the AI-driven explosion in DISTINCT designs (not wafer volume) turns mask-write throughput into the gate on design diversity -- remains absent from mainstream financial and market coverage.	Installed IMS multibeam-mask-writer write-hours per year at merchant and captive mask shops (Toppan, Tekscend, TSMC/Inte	72%	38%	2031-12-31

Vision P = strength of the structural case. Clause P = calibrated odds the exact dated clause resolves true, scored with Brier. The gap is the honest timing and measurement tax, not timidity.

P1 · The binding ceiling on the AI-compute wafer build-out is not ASML, EUV source power, or fab shells but Zeiss SMT ion-beam-figuring throughput for EUV/High-NA projection mirror sets at a single site in Oberkochen, Germany.

The boom: The loud narrative: ASML is the monopoly and EUV source power is the throughput knob. Capital is pouring into fab shells (TSMC AZ, Intel, Samsung, Rapidus) assuming scanners will follow. Sell-side equity models and export-control frameworks both treat the ASML scanner as the atomic unit of scarcity, without disaggregating the projection optics module as a distinct, slower-moving constraint. · Domain: semiconductors

DIRECTIONAL VISION

82%

STRICT CLAUSE

52%

RESOLVES

2032-12-31

An EUV scanner's resolution lives entirely in its Zeiss SMT projection optics: six aspheric Mo/Si multilayer mirrors polished and ion-beam-figured to roughly 50-picometer surface deviation, then coated with 50-plus individually sputtered Mo/Si layer pairs. Each mirror requires iterative measurement-correction cycles using custom interferometers that themselves take years to build and qualify. A full set is effectively a one-year artisanal build. This work is concentrated at Zeiss SMT Oberkochen, home to three decades of accumulated tacit skill in deterministic ion-beam figuring plus closed-loop metrology. High-NA (0.55 NA) strictly worsens the constraint: anamorphic mirrors of larger aperture, tighter tolerances, lower yield per attempt. No second qualified projection-optics figuring site exists. Zeiss has expanded globally for mask inspection (AIMS EUV 3.0), but mask inspection is not projection optics figuring and the two share no relevant tacit-skill base. The supply elasticity of this input is near zero on any horizon shorter than a decade. As every hyperscaler and nation-state tries to add leading-edge capacity simultaneously, they are all contending for the output of one optics shop's polishing benches, and that output grows at best in the low single digits per year. Fab shells, HBM, and packaging can be funded in parallel; projection mirror sets cannot. Web search confirms High-NA volume ramp remains in the low tens of systems per year through at least 2028, with no second figuring site announced as of June 2026.

WHY IT IS PRE-CONSENSUS

Consensus prices ASML-the-company as the monopoly and EUV source power as the throughput knob. Sell-side equity models do not disaggregate Zeiss optics-figuring throughput as a distinct capacity ceiling. Export-control frameworks (US BIS, Dutch Dekra controls) target the ASML scanner and DUV steppers; no control targets optics modules separately. Specialist semi-industry media (SemiconductorX, The Elec) mention the Zeiss dependency obliquely but frame it as addressed for standard EUV, without modeling it as the binding governor for the High-NA ramp. FUTURE_MAP covers EUV source-side constraints (InP CW lasers, dry resist) but not projection-optics fabrication throughput. The specific migration of the binding constraint from scanner to optics figuring at a single tacit-skill site is absent from equity modeling and policy framing.

HONEST PRICE CHANNEL

No direct financial instrument prices Zeiss optics-figuring throughput. ASML trades on order backlog and system shipments; the optics module is not disaggregated in any sell-side model. No second-source or capacity-expansion announcement has opened a price channel as of June 2026.

THE NEEDLE

Zeiss SMT deterministic ion-beam-figuring and Mo/Si multilayer-coating throughput for EUV/High-NA projection mirror sets at sub-100pm figure accuracy, single site Oberkochen, tacit-skill-bound, approximately one year per mirror set.

LEADING METRIC

Zeiss SMT / ASML disclosed or inferable annual EUV plus High-NA projection-optics module output (proxy: ASML EUV plus High-NA system shipments capped by optics, vs. ASML's stated build-capacity ambition). Track the gap between ASML order backlog and delivered EUV units attributable to optics lead time, and any Zeiss SMT capacity-expansion or second-figuring-site announcement. Current baseline: High-NA production rate in low tens per year through 2028 per industry projections; no second figuring site exists as of June 2026.

KILL-CRITERION

Kill if by 2032 Zeiss/ASML demonstrably break the one-set-per-year figuring rate at scale (for example a qualified second figuring or coating site outside Oberkochen, or robotic/computational-polishing pushing High-NA optics-module output above roughly 30 sets per year), OR if leading-edge wafer demand is met without optics being the binding scanner sub-component (source, stage, or throughput dominates the delivery gap instead), OR if a non-Zeiss EUV optics supplier qualifies for production scanners.

REFUTE CHECK (SURVIVED)

Three honest attacks. First: ASML management stated in 2024 that optics supply challenges from Zeiss had been resolved -- but that referred to standard EUV, and High-NA resets the difficulty at strictly tighter tolerances, so this is not a kill. Second: High-NA is early-ramp and not yet where the bulk of AI-accelerator wafers are made. The constraint bites hardest at sub-2nm nodes; standard EUV (0.33 NA) continues to serve 3nm/2nm where current AI training silicon lives. This is a real scope limitation and keeps clause_p from being high even though the physical mechanism is credible. Third: 6.5 years is long enough for partial supply response given enormous financial incentive; Zeiss is trying (roughly 2,000 of 9,349 employees working on High-NA as of 2025). Robotic and computational polishing are active research areas. The question is whether any of this translates to a qualified second figuring site or a step-change in throughput by 2032. The physical difficulty and tacit-knowledge moat make it unlikely on that timeline but not impossible. The structural claim survives because no current evidence contradicts the core physical inelasticity and no second site is on the horizon.

Why this call earned a place The physical mechanism is real and well-specified. The supply inelasticity at Oberkochen is genuine and not replicated elsewhere. The pre-consensus framing is materially wrong at the level that matters: equity and policy both treat the scanner as atomic when the actual constraint is one optics shop's polishing benches. The needle is precise, the kill criteria are honest, and the call survives adversarial attack. Scope risk (High-NA is not yet the dominant AI-wafer node) and partial-supply-response risk over 6.5 years keep clause_p at 0.52 rather than higher, but the structural vision_p is 0.82. Promote with the scope caveat marked.

P2 · Senior yield/process-integration engineers with multi-node sub-5nm ramp experience are the non-cloneable binding constraint on ex-Taiwan leading-edge output through the 2030s; capital and wafer capacity are financializable but a decade of hands-on ramp knowledge is not, and the rent migrates to whoever holds that headcount.

The boom: Capital, EUV tools, and wafer capacity are all fundable on a capex timeline; the one input that is not is a human who has personally climbed the yield curve across several node generations at sub-5nm class. Process integration and yield-enhancement engineering is overwhelmingly tacit: contamination fingerprinting, defectivity root-cause, and parametric drift diagnosis are learned over 10-plus years on a running line and do not transfer from a textbook or a two-year program. The market prices a fab as steel, tools, and a node label, then assumes yield ramp follows on schedule. It does not. TSMC explicitly attributed Arizona production delays to skilled worker shortages and responded by flying in over 1,000 Taiwanese engineers on three-year assignments -- then the second Arizona plant slipped to 2027-2028. That is not a capex problem; it is a headcount-of-experienced-engineers problem. One-third of the US senior fab-engineer cohort is at or near retirement age (a fixed birth-cohort fact), degree enrollment in feeder programs lagged for years, and announced ex-Taiwan fab count is contractually committed. The mismatch between capex-speed fab buildout and biology-speed expertise accumulation is mechanical and already locked in by decisions already made. · Domain: semiconductors

DIRECTIONAL VISION

82%

STRICT CLAUSE

58%

RESOLVES

2033-12-31

Demographic and learning-curve facts, not forecast of behavior. The experienced-engineer pool grows only at the rate a person accumulates a decade of on-line ramp experience; you cannot parallelize that with money. The senior US cohort retirement wave is a fixed birth-cohort event. The feeder-degree enrollment lag is already realized. The ex-Taiwan fab commitments (Arizona, Japan, Germany, CHIPS-funded US) are contractually binding. TSMC's own Arizona experience is a live proof: even with unlimited budget and top-brand pull, the solution was to import the experienced headcount, not to grow it locally fast.

WHY IT IS PRE-CONSENSUS

Consensus prices fabs as capex (tools, buildings, wafers) and treats yield ramp as a scheduled engineering deliverable that capital and CHIPS subsidies will deliver on time. Equity and policy models carry no separate line for whether enough experienced integration engineers exist to ramp a new line, and where workforce is mentioned at all it is framed as a generic technician or STEM-pipeline headcount story (trainable in two years), not as a decade-deep, tacit, non-cloneable senior-engineer constraint concentrated in one company in one country. The TSMC Arizona delays were covered in financial press as a worker-shortage story without any structural analysis of why the shortage is inelastic or what it implies for ex-Taiwan leading-edge yield timelines through the 2030s. Markets price the wafers; they do not price the wafer-curve carriers.

HONEST PRICE CHANNEL

No direct financial instrument prices yield-ramp speed per se. TSMC trades on wafer-price and utilization; Intel and Samsung on process-generation and capex. The possibility of structural ex-Taiwan output underdelivery driven by senior-engineer scarcity is not carried in sell-side models. AI-assisted process control (APC, ML defect detection, digital twins) is a real counter-force and should be watched, but current

evidence is that these tools amplify experienced engineers rather than replace their diagnostic judgment: TSMC Arizona hit production targets after importing experienced staff, not before.

THE NEEDLE

Senior (second-decade, 10-plus years) yield-enhancement and process-integration engineers with multi-node hands-on ramp experience at sub-5nm-class nodes. Not engineers in general, not technicians, not designers: the specific tacit-knowledge-bearing integration and yield headcount that determines how fast a new advanced line reaches qualified yield. Concentrated inside TSMC and in Taiwan; replaceable only at roughly one decade per head.

LEADING METRIC

(1) Time-to-qualified-yield (months from tool move-in to greater than 80 percent of mature yield) for ex-Taiwan leading-edge lines versus the Taiwan baseline for the same node; the gap stays above 12 months through the early 2030s. (2) Total compensation and signing premium for senior process-integration and yield engineers rising faster than fab construction cost indices. (3) Count of disclosed instances of TSMC, Intel, and Samsung relocating experienced Taiwanese or Korean engineers to staff ex-home-country ramps. (4) US senior fab-engineer vacancy duration and the SEMI/Deloitte engineer-shortfall figure trending upward, not closing, through 2030.

KILL-CRITERION

Kill the call if, through the early 2030s, ex-Taiwan leading-edge lines reach Taiwan-parity qualified yield within roughly the same time window as the Taiwan baseline (gap closes to under 6 months) without importing experienced engineers; OR if AI-assisted or automated yield-ramp and self-learning process control demonstrably substitute for senior integration headcount at scale, with a major fab publicly attributing a parity-speed ramp to automation rather than experienced staff; OR if senior process-integration engineer real compensation premiums flatten relative to construction-cost indices, indicating the scarcity dissolved.

REFUTE CHECK (SURVIVED)

Three challenges survive scrutiny but do not kill the call. First, AI-assisted process control could compress ramp times faster than expected; the counter is that TSMC Arizona achieved faster-than-expected ramp only after importing the experienced Taiwanese headcount, indicating automation is a multiplier on the scarce input rather than a substitute for it. Second, ex-Taiwan yield lags may be attributed to other causes (permitting, tariffs, political friction) rather than engineer scarcity in public disclosures, creating a measurement problem at resolution; this is a resolution-risk, not a structural refutation. Third, the general workforce-shortage story is known to industry practitioners and has appeared in trade and financial press; however, known-to-practitioners does not equal priced-in-equity-models, and no sell-side model carries the specific yield-ramp-lag-as-function-of-senior-headcount-density mechanism. The call survives all three.

Why this call earned a place The causal mechanism is correct and confirmed by live evidence (TSMC Arizona delays, engineer importation, second-plant pushout to 2027-2028). The inelastic input is correctly named at the right level of specificity. The demographic and learning-curve facts are locked in by decisions already made. The pricing gap is genuine: no financial instrument or equity model carries this constraint, and the surface worker-shortage narrative in press coverage does not price the structural rent-migration implication. Vision probability is high (0.82) because the mechanism is real and the evidence is direct. Clause probability is lower (0.58) because the primary metric (time-to-

qualified-yield versus Taiwan baseline) requires data TSMC does not publish, and AI-assisted ramp tools represent a legitimate technology counter that could narrow the gap faster than the structural argument implies.

P3 · Ruthenium becomes the gating input for advanced-logic interconnects as copper hits its resistivity wall at sub-2nm pitches, with supply hard-capped at roughly 30 tonnes per year as a platinum byproduct and no primary mine to develop.

The boom: Copper interconnects fail by physics at wire dimensions below roughly 12-16 nm pitch: electron surface scattering and grain-boundary scattering blow up resistivity, and the barrier/liner layer that copper requires eats an ever-larger fraction of the shrinking cross-section. Ruthenium is the industry-chosen escape because it can be deposited barrierless, its resistivity degrades far less at nanoscale, and it is being designed into the tightest metal levels and backside-power-delivery vias at the A14 node and below, with ramp roughly 2027-2030. The chokepoint is that ruthenium is not mined directly. It is a byproduct of platinum-group-metal extraction, roughly 30 tonnes per year globally, geologically concentrated in South Africa's Bushveld and Russia's Norilsk, with output governed by platinum and palladium demand from autocatalysts and jewelry, not chip fabs. When Ru moves from trace use in hard disks and catalysts to coating the bottom interconnect levels and backside vias of every advanced GPU, CPU and HBM die, even a few extra grams per wafer times tens of millions of leading-edge wafers collides with that 30-tonne ceiling. Rent migrates from the foundry to whoever controls Ru refining and sputtering-target conversion. · Domain: semiconductors

DIRECTIONAL VISION

72%

STRICT CLAUSE

40%

RESOLVES

2035-12-31

Ruthenium supply is geologically inelastic: it is a byproduct of PGM mining with no primary deposit, decade-plus mine lead times, and output set by platinum and palladium economics rather than chip demand. This is the same byproduct trap as iridium and helium-3. Substitution back to copper is foreclosed by electron scattering physics at sub-2nm pitches. Molybdenum is a partial competitor at the lowest levels but faces its own supply and deposition constraints and does not eliminate Ru demand. The demand step is mandatory and synchronized across the entire leading edge, not optional or vendor-specific.

WHY IT IS PRE-CONSENSUS

The structural supply-chokepoint framing (Ru as a 30 t/yr PGM byproduct conscripted as a mandatory input for all advanced logic) is absent from chip-supply-chain equity coverage at major banks and from fab-level public disclosures as of mid-2026. However, the metals commodity market has already partially moved: ruthenium exceeded \$1,700/oz by mid-March 2026 against a prior-year price near \$500/oz, meaning specialist metals research has noticed and partially priced the scarcity. The pre-consensus claim is therefore valid in the semiconductor supply-chain channel (not in chip-stock coverage or fab procurement disclosures) but is NOT valid in the metals market itself, where the price move has already been substantial. The pre-consensus window is narrower than the candidate asserts.

HONEST PRICE CHANNEL

Ruthenium spot has moved from roughly \$500/oz to over \$1,700/oz by mid-March 2026, a roughly 3x move, driven by scarcity and industrial demand including semiconductor applications. The Ru:Pt ratio has already inverted (Ru now above Pt in absolute price), partially satisfying the resolution metric before the A14 ramp begins. The call is partially priced in the metals market but not in semiconductor supply-chain equity coverage.

THE NEEDLE

Ruthenium metal supply and high-purity Ru sputtering-target and ALD-precursor conversion capacity for barrierless interconnect fill and backside-power vias at the A14 node and below. The inelastic node is the roughly 30 t/yr PGM-byproduct ruthenium stream plus target/precursor conversion, not the deposition tools or the transistor itself.

LEADING METRIC

(1) Ru:Pt price ratio versus its 2024-2026 baseline, with the call requiring at least a doubling. Note: ruthenium has already moved from roughly \$500/oz to over \$1,700/oz by mid-2026, partially satisfying this condition before A14 volume ramp even begins. (2) Estimated semiconductor share of annual ruthenium consumption, rising from low-single-digit percent toward a dominant share as A14-and-below ramp. (3) Public citations by TSMC, Intel, Samsung or their materials suppliers naming ruthenium supply or Ru sputtering-target availability as an interconnect or backside-power gating item.

KILL-CRITERION

Killed if by 2035-12-31: (a) the industry extends copper or adopts an abundant alternative such as molybdenum, graphene caps or air-gap schemes at the tightest levels such that ruthenium never becomes the dominant bottom-level or backside metal in volume, OR (b) ruthenium recycling plus modest PGM-supply growth keeps the Ru:Pt ratio within roughly 50% of its 2024-2026 baseline despite semiconductor adoption, OR (c) sub-2nm and CFET logic stalls so the volume of Ru-dependent wafers never reaches the scale that strains the 30 t/yr ceiling.

REFUTE CHECK (SURVIVED)

Three live attack vectors. First and most material: the Ru price has already tripled from its prior-year level by mid-2026, before A14 volume ramp has started. This means either the price has already discounted forward demand (reducing remaining upside and the clause resolution odds) or the move was driven by non-semiconductor factors and the chip-demand shock is still coming. Either way, the clean pre-consensus framing is compromised. Second: molybdenum is a documented alternative at the tightest metal levels. If Mo takes the bottom-level fill role and Ru is used only for liners, per-wafer Ru mass drops significantly and total semiconductor demand may stay within supply growth from recycling and scrap recovery at higher prices. Third: A14 volume ramp timing is uncertain. TSMC's A14 is a 2027-2028 earliest volume target and CFET yield risks could delay meaningful wafer volumes into the early 2030s, compressing the window before the 2035 resolution date. The structural mechanism survives all three attacks but the clause probability is materially lower than the vision probability because of partial prior pricing, Mo substitution risk, and ramp timing uncertainty.

Why this call earned a place Promoted because the physical forcing mechanism is sound and verified (copper resistivity wall at sub-2nm, Ru the chosen replacement), the supply inelasticity is genuine and confirmed (PGM byproduct, no primary mine, decade-plus lead times), and the semiconductor supply-chain angle remains unpriced in chip-stock equity coverage even as the metals price has moved. The partial prior price move is a real headwind to the pre-consensus claim but does not kill the call: the chip-supply-chain channel has not yet absorbed this as a fab-ramp gating risk, which is where the remaining edge sits. Molybdenum substitution and ramp timing are the live kill risks and keep clause_p well below vision_p.

P4 · The binding constraint on the US/EU fab buildout is not capital or tools but the tiny, slow-to-certify pool of UHP process-piping welders and hookup fitters; their scarcity makes the announced 2023-2030 leading-edge capacity schedule structurally undeliverable.

The boom: A fab is two things: a building shell and an extraordinarily complex web of ultra-high-purity gas, chemical, slurry, and process-water distribution that must be orbital-welded, hooked up, and certified leak-tight to SEMI particle standards before a single tool runs. That hookup work is a distinct specialty trade, certifiable only after years of pipefitting experience plus fab-specific qualification to smooth-bore, crevice-free, helium-leak-tested standards. TSMC Arizona already slipped explicitly citing this: the company had to airlift roughly 500 Taiwanese hookup specialists because the regional certified pool was too thin for even one major fab. SEMI's own data show roughly 97 high-volume fabs launching globally in 2023-2025, with the largest US/EU share in history and all on overlapping schedules. The concurrent demand is a step-function; the certified-trade pool grows only at the rate the trades can train and qualify people, which is years. Rent migrates to the specialty UHP mechanical subcontractors and to the certified orbital-welder pool; the visible symptom is schedule slip clustered on the install/hookup phase, not on construction or tool delivery. · Domain: semiconductors

DIRECTIONAL VISION

72%

STRICT CLAUSE

52%

RESOLVES

2031-12-31

Labor-supply and certification-time arithmetic. The pool of UHP-qualified welders and hookup fitters in any region grows only as fast as the trades can train and certify people, on a multi-year timeline, while demand is a step-function from contractually committed, near-simultaneous fab groundbreakings. Prefabrication and modular UHP skidding shift some work off-site but do not eliminate the on-site qualified-hookup labor requirement: certified fitters are still required for final connections, leak testing, and tool hookup under SEMI standards. Robotic orbital welding systems reduce headcount per weld but still require a qualified operator for setup, calibration, and per-weld certification; they blunt but do not dissolve the constraint. The TSMC AZ response (imported Taiwanese crews, emergency Local 469 partnership) is the direct empirical signature of genuine supply inelasticity. UHP tubing installation costs rose approximately 18% between 2022 and 2024, consistent with a constrained specialty-trade market.

WHY IT IS PRE-CONSENSUS

Consensus frames the fab buildout as a capital and policy story: CHIPS dollars committed, tools on order, ground broken, therefore capacity arrives on schedule. Where labor is mentioned, it is the generic construction-worker-shortage framing. The TSMC Arizona delay was covered in trade press but read as a one-off local labor dispute, not as a structural, replicating constraint across every concurrent project drawing from the same regional pool. No sell-side equity coverage separately prices the specialty UHP mechanical subcontractor tier or the certified-orbital-welder pool as a distinct critical-path risk. The specific claim -- that the binding constraint on the entire announced ex-Asia fab map is a few thousand certified UHP tradespeople per region, and that the slip will cluster on the boring, invisible tool-hookup phase that no equity model carries as a line item -- is not in the analytical consensus. The demand-softening escape valve (Samsung Taylor pause, Intel Germany reassessment) is the main path by which the clause fails to resolve as stated even if the structural mechanism is real.

HONEST PRICE CHANNEL

Not separately priced. Trade-press coverage of the TSMC AZ delay is public, but it is framed as a one-off workforce story, not as a structural node that replicates across the concurrent US/EU fab wave. The specialty UHP mechanical subcontractor tier is not a distinct line item in any equity model or policy analysis I can find. The 18% install-cost escalation in UHP tubing systems (2022-2024) is a billing-rate signal that has not been translated into fab-schedule risk pricing. Narrative presence does not equal price-channel pricing here.

THE NEEDLE

Certified UHP orbital welders and UHP/PVDF gas-and-chemical-distribution fitters qualified to semiconductor contamination standards, specifically the billable trade-hours of this pool per concurrent fab project. Not general construction labor, not electricians broadly, not the building shell: the specific UHP install-and-hookup trade hours that gate the transition from a built fab to a tool-ready fab.

LEADING METRIC

(1) Share of announced US/EU leading-edge fab projects whose first-production date slips, with the install/hookup or specialized-equipment-install phase named as the cited cause versus tool delivery, demand, or financing. (2) Union UHP/pipe-trade journeyman wage and total-package escalation in fab-cluster locals (Arizona pipe trades, Ohio, Texas) relative to the broader construction wage index. (3) Specialty UHP/mechanical subcontractor backlog and billing-rate trends, and frequency of imported Taiwanese or Korean hookup crews on US sites. (4) Count of CHIPS-funded leading-edge fabs reaching qualified first production on or after their originally announced date, with skilled-trade hookup labor cited.

KILL-CRITERION

Kill if the bulk of announced US/EU leading-edge fabs reach qualified first production on or near their originally announced schedules without hookup/install labor being cited as a binding cause. Kill if prefabrication/modular UHP skidding plus robotic orbital welding demonstrably collapse on-site UHP hookup labor demand such that a major fab publicly attributes an on-time hookup specifically to off-site modularization displacing the scarce on-site trade. Kill if UHP pipe-trade wage and billing-rate escalation in fab clusters flattens to the general construction index, signaling the scarcity cleared. Kill if leading-edge demand softening delays the majority of the concurrent fab wave for demand or financing reasons before hookup labor can become the cited gating phase.

REFUTE CHECK (SURVIVED)

Strongest attacks: (1) Already public -- the TSMC AZ story ran in 2023 and is well-covered. Survives because trade-press coverage at the one-fab level is not the same as the replication thesis being priced across eight to twelve concurrent projects. (2) Automation -- robotic orbital welding exists. Survives because the machine still requires a qualified operator for per-weld setup and SEMI-grade certification; headcount reduction is real but the trade-pool constraint is not dissolved. (3) Demand softening kills the clause before hookup labor matters -- this is the live risk that keeps clause_p below vision_p. Samsung Taylor is already paused for demand reasons; if leading-edge demand softens enough, fab schedules slip for demand causes and hookup labor never becomes the stated binding constraint even if it would have been one. The structural mechanism is genuine; the clause resolution is uncertain because the demand-softening escape valve is non-trivial.

Why this call earned a place The structural case is clean: inelastic supply (years to certify), step-function demand (near-simultaneous committed groundbreakings), and a live empirical signature

(TSMC AZ imported 500 specialists, Local 469 emergency partnership, 18% install-cost escalation). The pre-consensus gap is real: the specific UHP hookup-trade-pool framing is absent from analytical coverage and equity pricing even though the one-off TSMC story is public. The call survives the automation and already-priced attacks. The main uncertainty that separates clause_p from vision_p is the demand-softening path: if leading-edge demand slows materially before 2028, the concurrent fab wave shrinks and hookup labor ceases to be the cited gating constraint even if the structural bottleneck remains latent.

P5 · Enriched Si-28 isotope-separation cascade capacity becomes the gating substrate input for silicon-spin qubits and (if adopted) 3D-logic thermal relief, with world supply sitting at tens to low hundreds of kg/yr against a potential tons-per-year demand cliff by the early 2030s.

The boom: Two independent demand curves land on one isotope-separation step. Silicon-spin qubits from Intel, Diraq, Quantum Motion and peers require 99.99%+ Si-28 to suppress Si-29 nuclear-spin decoherence -- this is non-negotiable physics, not a preference. Separately, removing Si-29 and Si-30 from the silicon lattice boosts thermal conductivity 60-600% over natural silicon, which matters specifically for buried-device heat extraction in CFET and backside-power-delivery 3D stacks where junction temperatures are the binding physical limit. The enrichment process -- aerodynamic separation of silane or gas-centrifuge on SiF₄ -- is the same physics as uranium enrichment, so it carries export controls, long permitting timelines, and high capital intensity. ASP Isotopes (ASPI) is the only disclosed Western commercial enriched-Si-28 supplier, with a Pretoria facility restarted in May 2026 after engineering fixes, first commercial shipments targeted Q3 2026, and capacity documented at greater than 80 kg/yr from one facility. Three named purchase agreements already exist on that output (a major U.S. semiconductor company, a large industrial gases company, a large U.S. buyer). Total world capacity including any Rosatom legacy output is on the order of low hundreds of kg/yr. Any scenario where silicon-spin qubits reach even modest commercial volume, let alone where Si-28 epi gets adopted for 3D-logic thermal relief, requires 10 to 100 times that output. Aerodynamic and centrifuge cascades are among the slowest industrial plants to permit and commission; \$333M in ASPI cash begins to relax supply on a multi-year basis but does not dissolve the constraint inside a decade. · Domain: ai-compute

DIRECTIONAL VISION

72%

STRICT CLAUSE

34%

RESOLVES

2034-12-31

Isotope-separation cascade capacity for a chemically identical, one-neutron-difference isotope is physically inelastic on sub-decade timescales. You cannot demand-pull tons of Si-28 the way you can ramp a fab: the equipment is export-controlled, the cascades take years to commission, and there is no substitution (Si-28 is defined by the absence of Si-29 spin and the mass difference from Si-30). The inelastic node is separative work units for silicon isotopes, not the downstream wafer or fab process. Two demand vectors -- spin-qubit substrates and 3D-logic thermal channels -- converge on the same separation step, and neither is served by recycling or a materials workaround.

WHY IT IS PRE-CONSENSUS

ASP Isotopes (ASPI) is a public equity already being bought on the quantum-substrate thesis, and \$333M has been raised against it, so the story is not invisible. However, the consensus framing treats ASPI as a niche quantum-materials micro-cap, not as a gating input to semiconductor foundry economics comparable in structural importance to photoresist or specialty gases. The dual-demand convergence -- that the same export-controlled separation step gates both a qubit winner and a 3D-logic thermal-relief path -- does not appear in mainstream semiconductor sell-side coverage or foundry cost models. The magnitude of the supply-to-demand gap (10 to 100x scale-up required) is not reflected in any public analyst model I can identify. Pre-consensus in the consequential sense: the equity exists, but the structural semiconductor-input framing does not. Borderline, but survives the price-channel check because there is no liquid spot market and no sell-side price target on enriched-Si-28 feedstock itself.

HONEST PRICE CHANNEL

No public spot or contract price for enriched Si-28 feedstock is disclosed. ASPI does not publish contract pricing. Natural electronic-grade polysilicon trades around \$5 to \$15/kg; enriched Si-28 at research-grade quantities is reported in academic literature at roughly \$1,000 to \$10,000/kg, implying a current multiple of 100x to 1,000x, but this is not a liquid market price. The absence of a public price channel is itself a signal that the market is pre-commercial and not yet arbitrated. No sell-side price deck exists. The resolve condition references a greater than 5x multiple, which is almost certainly already exceeded in any disclosed transaction -- the risk is whether a public, sustained, verifiable price benchmark exists by 2034, not whether the multiple is high.

THE NEEDLE

Enriched Si-28 isotope-separation capacity: specifically the aerodynamic-separation (silane) or gas-centrifuge (SiF₄) cascade throughput that converts natural silicon into 99.99%+ Si-28 feedstock. This is the inelastic node, not the wafer fab, not the qubit design, not the epi reactor.

LEADING METRIC

(1) Global installed enriched-Si-28 (99.99%+) production capacity in kg/yr across all suppliers (ASP Isotopes/Pretoria, any Rosatom/legacy Russian output, new entrants) -- baseline today is on the order of tens to low hundreds of kg/yr from a literal handful of facilities. (2) Contract or spot price of enriched Si-28 feedstock or enriched silane as a multiple of natural electronic-grade polysilicon. (3) Count of commercial (non-R&D) silicon-spin-qubit or Si-28-epi-logic programs that publicly identify enriched-Si-28 substrate supply (not qubit design, not fab access) as a named procurement gate. Resolves TRUE if by 2034-12-31 enriched-Si-28 supply is publicly cited as a binding gate by at least two commercial programs AND enriched-Si-28 feedstock sustains a greater than 5x price multiple over natural electronic-grade silicon.

KILL-CRITERION

Killed if by 2034-12-31 any of: (a) world enriched-Si-28 capacity scales past roughly 1 ton/yr with the price multiple compressing below 3x, meaning supply caught up to demand; (b) silicon-spin qubits lose the modality race to superconducting, trapped-ion, or neutral-atom AND Si-28 epi is never designed into volume 3D-logic, collapsing both demand legs simultaneously; (c) a non-cascade enrichment route -- laser isotope separation, plasma separation, or sufficiently spin-clean natural-Si error correction -- reaches industrial scale and dissolves the separation chokepoint.

REFUTE CHECK (SURVIVED)

Three genuine challenges. First: ASPI is already public and \$333M in cash means scale-up investment is happening, which begins to relax supply over an 8-year horizon. The supply inelasticity is real today but may erode faster than the clause assumes if ASPI or a competitor adds cascade stages aggressively. Second: the 3D-logic demand leg (Si-28 epi for CFET thermal relief) is technically sound but speculative as a volume design-in. No major foundry (TSMC, Samsung, Intel Foundry) has publicly committed to Si-28 epi channels for volume CFET production. If this leg stays a research curiosity, total addressable demand is smaller and the supply gap may close on a longer timeline without becoming a binding gate. Third: qubit-modality risk is real over 8 years. Superconducting qubits (IBM, Google) and trapped-ion (IonQ, Quantinuum) are well-funded competing paths; if silicon-spin does not win commercial volume by the late 2020s, the quantum demand leg weakens and the clause may not resolve even if the structural mechanism is correct. The candidate survives on mechanism and supply inelasticity, but clause probability is meaningfully suppressed by these three factors.

Why this call earned a place The physical mechanism is sound and the inelastic needle is correctly identified. Isotope-separation cascade capacity for Si-28 is genuinely inelastic on sub-decade timescales, the supply gap is large and real, and the dual-demand convergence (qubits plus 3D-logic thermal) is structurally original relative to consensus coverage. Vision probability is high because the constraint is real. Clause probability is held materially lower: \$333M in active investment partially relaxes supply over 8 years, the 3D-logic demand leg is unconfirmed by any foundry design-in, qubit-modality competition is unresolved, and the resolve condition's price-multiple verification depends on a public market that does not yet exist. This single Si-28 framing absorbs and supersedes a near-duplicate candidate (same needle, lower scores).

P6 · Multibeam mask-write throughput (IMS Nanofabrication, sole production vendor) becomes the binding gate on how many distinct High-NA AI-chip designs the industry can tape out per year by 2031.

The boom: The loud narrative is the GPU and ASIC design explosion -- design talent, HBM, CoWoS packaging. Masks are treated as a solved commodity. The photomask conversation, where it exists, is about blank substrates (Hoya, AGC), not about who writes the pattern. The specialist trade press (SemiAnalysis, August 2022) has named the IMS monopoly, but the second-order mechanism -- that the AI-driven explosion in DISTINCT designs (not wafer volume) turns mask-write throughput into the gate on design diversity -- remains absent from mainstream financial and market coverage. · Domain: semiconductors

DIRECTIONAL VISION

72%

STRICT CLAUSE

38%

RESOLVES

2031-12-31

High-NA EUV mandates curvilinear inverse-lithography-technology masks: sub-2nm-class features require pre-distorted freeform patterns that fragment into millions of shots on legacy variable-shaped-beam writers, making write times economically ruinous. Only multibeam mask writers, which are shape-agnostic and write any pattern in roughly constant time (7-12 hours per layer), make curvilinear ILT viable at scale. Exactly one multibeam architecture has ever reached production: IMS Nanofabrication (Vienna), Intel-acquired 2015, with roughly 50 tools in the field. The AI-ASIC boom multiplies the count of DISTINCT designs and thus the count of unique mask sets (60-80+ layers each, many EUV). Distinct-design count is rising far faster than the installed base of IMS writers can grow: these are precision e-beam instruments with 18-36 month lead times and low annual unit volume. The binding constraint on design diversity migrates from scanner wafer-throughput (which serves high-volume parts well) to mask-write throughput, which scales with distinct design count and is bottlenecked on a single-vendor tool base. Important update: Intel has sold minority stakes to TSMC (~10%), JEOL (~2.5%), and Bain Capital (~20%) at a ~\$4.3B valuation (2023). The pure competitor-controlled-chokepoint framing is partially defused, but the sole-source supply constraint on the physical tool is unaffected by ownership structure.

WHY IT IS PRE-CONSENSUS

The IMS monopoly as a static fact was named by SemiAnalysis in August 2022 and is known in the specialist semiconductor investor community. What is not priced: the second-order mechanism that the AI-ASIC distinct-design explosion (not wafer volume) is the demand driver that converts a known monopoly into the active binding throughput constraint. The ownership structure has also shifted materially (TSMC, JEOL, Bain now co-holders), which defuses but does not dissolve the supply inelasticity argument.

HONEST PRICE CHANNEL

SemiAnalysis August 2022 piece named the IMS monopoly explicitly. Intel minority-stake sales (TSMC, JEOL, Bain) are public press releases with a \$4.3B valuation. The static monopoly fact is in the specialist price channel. The demand-mechanism (distinct AI-design count as throughput driver) is not explicitly covered in equity research found in this search.

THE NEEDLE

Installed IMS multibeam-mask-writer write-hours per year at merchant and captive mask shops (Toppan, Tekscend, TSMC/Intel/Samsung in-house) versus the annual count of new leading-edge (EUV-layer) tape-

outs demanding curvilinear ILT masks. Proxy: quoted leading-edge mask-set lead times and price trends at merchant shops; any mask shop citing writer capacity as limiting tape-out cadence.

LEADING METRIC

Roughly 50 IMS MBMW tools in the field as of 2025, MBMW-301/401 class ramping for 3nm and below. Write time 7-12 hours per layer; 60-80+ layer sets per tape-out. No competing production multibeam writer qualified at any leading mask shop. IMS valued at ~\$4.3B in 2023 minority-stake transactions. Europe's first MBMW installed at Tekscend November 2024.

KILL-CRITERION

Kill if by 2031: (1) a second independent production multibeam mask-writer vendor qualifies at a leading mask shop (JEOL's minority stake and e-beam expertise makes this the most plausible path, but no competing tool is currently qualified); (2) Intel or the IMS consortium demonstrates structurally neutral allocation such that no customer faces competitor-controlled rationing; (3) leading-edge mask-set lead times stay flat and no mask shop cites writer capacity as a tape-out bottleneck; or (4) High-NA adoption stalls and curvilinear ILT never becomes the dominant leading-edge mask mode.

REFUTE CHECK (SURVIVED)

Three challenges survive. First, the competitor-owned framing is now stale: TSMC (the primary customer) and JEOL (a plausible future competing-tool developer) hold minority stakes, giving the industry co-governance levers. Second, SemiAnalysis coverage means the monopoly is partially in the specialist price channel even if it has not reached mainstream financial coverage. Third, the clause requires mask-write throughput to become categorically the gate by 2031 -- a strong claim when scanner throughput, CoWoS, and HBM remain competing candidates for binding constraint, and the industry has five years plus strong incentive to add capacity. What survives refutation: the physical mechanism is real, the installed base is thin (~50 tools for global leading-edge demand), lead times on capital equipment of this class are 18-36 months minimum, and no competing production tool exists today. The throughput math is structurally sound. The vision is solid; the exact clause fires at roughly even odds given the five-year horizon and active kill-condition paths.

Why this call earned a place The physical mechanism is real and the supply constraint is genuinely inelastic today. The pre-consensus check is partial: the static monopoly is known in specialist circles but the demand-side mechanism (distinct AI-chip design count, not wafer volume, as the throughput driver) is not explicitly priced. The ownership update (TSMC/JEOL/Bain minority stakes at \$4.3B) is material -- it partially defuses the competitor-allocation-chokepoint framing and represents a kill-condition path for the clause. Promote because the structural case is strong and the needle is physically specific, but set clause_p well below vision_p to reflect the active kill paths and the five-year window in which the industry can respond.

Seeds considered and not promoted

Cleared the physical-constraint test but failed on investability or on the price channel. Logged because the discipline is to surface what was cut.

Seed	Physical case	Why not promoted
By 2035, isotopically enriched Si-28 (SiF ₄ separation cascade throughput) is the binding supply constraint on both silicon spin-qubit fault-tolerant quantum machines and thermally-limited high-density 3D logic	Si-28 enrichment cascade, dual quantum + 3D-logic demand	Near-duplicate of P5 (same inelastic needle, same dual-demand mechanism, same sole Western producer) but at lower scores (vision 0.62 vs 0.72, clause 0.18 vs 0.34) and a 2035 vs 2034 horizon. P5 is the stronger framing of the identical call; keeping both would double-count one bet.
By 2032 the binding constraint on high-stack HBM and 3D-stacked logic is not bonding or substrates but the supply of ultra-low-alpha (ULA) lead and tin for bumps, microbumps and TSV fill	Ultra-low-alpha lead/tin radiopurity for HBM/3D-logic solder	Genuinely novel mechanism, but the lowest clause_p in the set (0.18) because the primary mitigation (Cu-Cu hybrid bonding) is already funded at exactly the cell-adjacent interface the thesis depends on, and the resolution trigger leans on a public supplier disclosure that may never appear even if the constraint is real. Highest dissolution risk of the eight; held as the first reserve.

Generated by the Pope System. Each call is a forward instrument: resolution date and kill-criterion fixed at creation, superseded never edited, clause probability scored with Brier at resolution.